

A Multimodal System for Empathy Expression: Impact of Haptic and Auditory Stimuli

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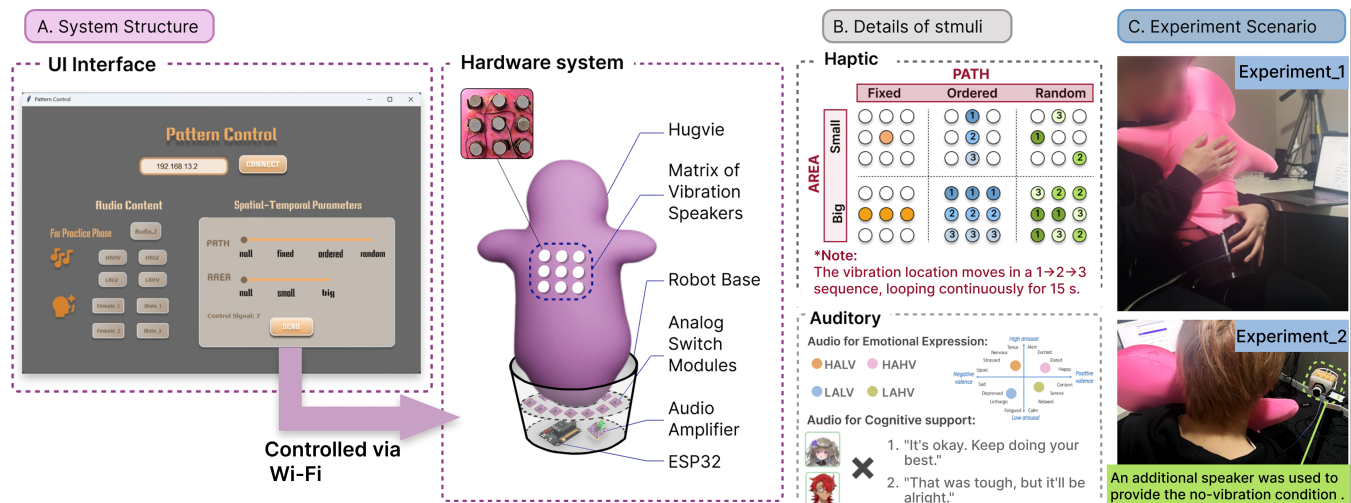


Figure 1: Design and development of the system. (A. The system is developed based on vibration speakers, controlled through an interface via WiFi, and capable of providing multimodal stimuli. B. Provides details of stimulus design for emotional transmission and cognitive support task. C. Highlights the user experience and experimental environment.)

Abstract

Empathy plays a crucial role in Human-Robot Interaction (HRI), enhancing emotional transmission and interaction experiences while breaking psychological barriers. Although many HRI systems focus on empathy expression, few have investigated the combined effects of spatial haptic parameters and auditory cues on emotional and cognitive empathy. To address this gap, we developed a multimodal HRI system based on a huggable robot. It is integrated with spatial haptic patterns and auditory cues. Two exploratory experiments were conducted to evaluate the effectiveness of these parameters in emotional transmission and cognitive support. Results revealed that

spatial haptic parameters primarily influenced arousal, while music significantly impacted valence. Synergistic effects between haptic and auditory modalities were observed under specific conditions. For cognitive support, participants preferred vibration combined with voice over voice-only feedback, while random haptic paths performed poorly across the cognitive dimensions. These findings suggest the potential of multimodal feedback in improving empathetic HRI systems.

CCS Concepts

• Human-centered computing → Sound-based input / output; Haptic devices; Empirical studies in HCI.

Keywords

empathy expression, interaction design, huggable multimodal system, haptic and auditory interaction, user studies

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1 INTRODUCTION

Integrating the concept of empathy into Human-Robot Interaction (HRI) systems is crucial for enhancing their interaction with humans [17]. Empathy has been recognized as a key factor in enhancing emotional transmission and interaction experiences, as it plays a central role in human emotional interaction by fostering interpersonal connections and breaking psychological barriers between the self and others [19].

The HRI systems occupied with empathy ability are supposed to provide effective emotional and cognitive support in various scenarios by responding appropriately according to human emotions, thoughts, and contexts [18]. Such empathetic responses include parallel responses that match others' emotions [14, 23], i.e. emotional empathy, and inferring others' feelings and needs and providing supportive responses [21], i.e. cognitive empathy. When these responses are translated into robots' outputs, they are termed empathy expressions, reflecting how robots demonstrate empathy in interactions.

Many existing studies have utilized various sensory modalities to achieve empathy expression. Touch is an intimate and emotionally rich sense that plays a critical role in emotional expression [5]. Prior research has primarily focused on non-spatial parameters like vibration intensity to simulate valence and arousal (Russell, 1980) [25] or on spatial parameters such as tactile stroking speed [28]. Although [7] focused on users' ability to perceive shapes and motion directions through touch, they have not explored how these spatial elements relate to empathy expression.

Sound, like touch, is a vital medium for emotional communication and cognitive support, widely utilized in empathy expression through music [13] or language [10]. Combining music with vibration-based haptic has demonstrated potential for transmitting emotional characteristics through the touch channel [22]. However, there remains a lack of systematic exploration on how these multimodal stimuli influence both emotional expression and cognitive empathy.

Taking inspiration from the aforementioned studies, using a multimodal system based on a huggable robot, we explored both the interaction effects of touch and sound as well as their individual contributions. The system utilizes a matrix of vibration speakers to enable variations in haptic spatial parameters and auditory cues, aiming to deliver multimodal emotional stimuli and cognitive support expressions. Through two exploratory experiments, we examined its role in emotional transmission and cognitive support. Therefore, the contributions of this paper are as follows:

- (1) Designed and developed a huggable robot system that integrates spatial haptic parameters and auditory cues to improve empathy expression.
- (2) Identified the effects of spatial haptic parameters and auditory cues on emotional expression and cognitive empathy.

These findings provide nuanced design insights for multimodal emotional support systems, focusing on the distinct roles of each modality.

2 DESIGN EXPLORATION

In this section, we present the system design process, including its inspiration, development, and the design details of multimodal stimuli.

2.1 Design Consideration

Hugging is a common way to express empathy and emotional support [11]. Although existing research on huggable robots focuses on tactile perception during hugging actions [3, 16], the multimodal interaction potential of chest contact remains underexplored. In real hugs, people often perceive the heartbeat, voice, and vibrations of the other person through chest contact. It shows the potential of integrating haptic and auditory stimuli through a huggable system.

While pressure and stroking have been used for emotional expression [8, 27], vibration can more directly simulate physiological activities such as heartbeat and vocal cord vibrations, which are perceived through both tactile and auditory channels during a hug. Therefore, this study employs vibration as the haptic stimuli to more naturally extend the authentic hugging experience and explore its multimodal interaction potential in empathy expression.

Previous studies have shown that moving haptic stimuli can effectively enhance arousal levels and that users can perceive complex spatial information, such as shapes and motion directions [26]. However, the specific role of haptic spatial parameters in emotional expression and cognitive support remains underexplored. This study focuses on two fundamental variables in haptic spatial patterns: vibration path and activation area, to examine their impact on multimodal empathy expression.

The auditory design of this study consists of two parts. For emotional expression, materials were selected from the DEAM dataset [1], categorized into four quadrants based on Russell's circumplex model affect [20]. This dataset was chosen for its rich emotional labels directly related to arousal and valence, widely used in emotion research. For cognitive empathy, stimuli were designed with equal duration and neutral content to reduce emotional bias from language and avoid the potential confounding effects of duration differences, ensuring that the focus remains on the effects of multimodal stimuli.

The system utilizes vibration speakers capable of delivering tactile and auditory feedback. This design ensures seamless integration of multimodal stimuli in a compact form factor. Our primary goal is to examine how spatial haptic parameters (path and area) and different auditory stimuli contribute to empathy expression and cognitive support, and to explore the potential of multiparameter control in these processes.

2.2 Development of the System

Inspired by the above considerations, we developed a multimodal system based on Hugvie [15, 24] to help participants engage with and understand empathy scenarios. It is controlled in real-time by software to provide synchronized haptic and auditory feedback (Fig.1.A).

Hardware architecture. The system utilizes a 3×3 vibration speaker matrix to achieve synchronized haptic and auditory feedback. Each 4W 3Ω vibration speaker (Almencla 27mm Model, UPC: 614089886358) is connected to the same ESP32 through an analog switch, allowing selective control to generate diverse haptic patterns. The speakers are also connected to the same amplifier (MAX98357) to ensure that activated speakers play the same audio signal, maintaining a consistent phase and intensity. Since the vibration is directly driven by the speaker, the frequency characteristics of the tactile stimuli directly inherit the spectral properties of the input audio signal. To ensure consistent tactile and auditory output levels, all audio inputs were normalized. While the sound pressure level was not directly measured, the airborne sound from the speaker was perceptible but did not interfere with the tactile experience, as confirmed by participant feedback. Most sensors are integrated in the base and speakers are covered with nylon fabric to protect the device and enhance the haptic experience.

Software implementation. The control interface communicates with the ESP32 via Wi-Fi, allowing the research assistant to select audio signals and adjust the haptic patterns. After sending the command, the audio signal and the haptic parameters are synchronized and transmitted to the ESP32, causing the system to provide multimodal stimuli in real time.

2.3 Stimuli Design

Haptic Stimuli. The combination of two parameters, movement path (Path) and activation area (Area), is shown in Fig.1.B.

- Path: Includes fixed (vibrating at fixed position), ordered (vertical shifting vibration), and random (altering the vibration position randomly). To enhance perceptual continuity and naturalness, all modes, except the fixed condition, switch locations every 500 ms.
- Area: Includes big (three speakers activated simultaneously) and small (one speaker activated at a time), used to study the impact of coverage on haptic experience.

Auditory Stimuli. Different auditory stimuli, shown in Fig.1.B, were designed to meet the requirements of two tasks. Emotional Empathy Task: Four 15-second low-to mid-frequency instrumental music clips were selected from the DEAM dataset through Fourier and cluster analysis, corresponding to the four quadrants of Arousal-Valence. Furthermore, a baseline music clip with emotional characteristics close to the mean of the dataset (Arousal = 4.8, Valence = 5.0) was selected for emotional calibration:

- HAHV (High Arousal, High Valence): Arousal = 6.0, Valence = 6.8
- HALV (High Arousal, Low Valence): Arousal = 6.83, Valence = 3.17
- LALV (Low Arousal, Low Valence): Arousal = 2.9, Valence = 1.9
- LAHV (Low Arousal, High Valence): Arousal = 4.4, Valence = 5.6

Cognitive Empathy Task: To simulate short-term cognitive support expression, two neutral encouragement phrases were designed: “It’s okay. Keep doing your best.” and “That was tough, but it’ll be alright.” Each phrase was generated via VOICEVOX, with 2-second

male and female voice versions to ensure diversity and reduce content bias.

All audio files mentioned were normalized to a maximum amplitude of -10dB using Audacity, ensuring that no change to the spectral characteristics or emotional content. This was done to maintain consistent vibration intensity across multiple speakers and avoid variations in tactile strength due to volume differences.

3 EXPLORATORY STUDY

3.1 Participants

We recruited 12 participants (6 males; 6 females; age: 22–63, $M = 36.25$, $SD = 13.9$), with none reporting hearing impairments. Of these, nine participants (5 males, 4 females; age: 22–63, $M = 35.44$, $SD = 15.50$) completed the second experiment, with three excluded due to system issues. As the experiments are independent, this does not affect the results. Pre-experiment questionnaires showed familiarity with haptic technology (e.g. vibration wristbands) and comfort with haptic stimuli. Most participants had strong musical understanding, except for one with low confidence.

Furthermore, the scores of the participants on the IRI-J scale [9] were as follows: Fantasy ($M = 2.99$, $SD = 0.81$), Perspective taking ($M = 3.06$, $SD = 0.79$), Empathic concern ($M = 2.72$, $SD = 0.70$) and Personal stress ($M = 2.32$, $SD = 0.47$). These results indicate moderate empathic abilities and emotional stability, making participants well suited to study empathic expression to multimodal stimuli.

3.2 Procedure

This study conducted two exploratory experiments to investigate: (1) the effects of spatial haptic parameters and emotional music on emotional transmission, and (2) the comparative impact of single-modal vocal stimuli and multimodal stimuli (combining haptic path patterns and vocal gender) on the effectiveness of support in cognitive empathy scenarios. Both experiments were conducted in a quiet environment, where participants sat in a chair while holding the huggable system, ensuring that all speakers of the device maintain consistent contact with their body.

3.2.1 Experiment 1: Focus on emotional transmission. Prior to the first experiment, participants completed a 10-minute familiarization session to calibrate emotional concepts and reduce cognitive load. They experienced baseline and emotional music as well as multimodal training before starting the formal experiment. The haptic patterns and emotional music used in the training were identical to those in the formal experiment. Following each stimulus, participants used a laptop to assess valence and arousal levels via Affective Slider (AS) [2], a numeric self-report tool with two slider controls. They rated: “To what extent do you perceive the emotion expressed by this pattern as negative or positive?” (Valence) and “To what extent does this pattern convey a sense of high or low arousal (active or calm)?” (Arousal). These questions ensured that the assessment focused on participants’ perception of the system’s emotional expressions rather than their own emotional states. The stimuli were remotely activated by a research assistant. Participants were exposed to 24 randomized stimuli (combinations of emotional music and haptic patterns, as detailed in Section 2.3). The stimuli

were randomized to avoid order bias and fatigue effects. Each stimulus was separated by a 25-second interval, allowing enough time for participants to complete their ratings and return to their baseline emotional state before the next stimulus. The entire session lasted approximately 30 minutes. After the experiment, participants participated in a brief semi-structured interview to provide feedback on their understanding of multimodal stimuli, their perspectives on their role in emotional transmission, and suggestions for improvement.

3.2.2 Experiment 2: Focus on Cognitive Support. Participants were exposed to a simulated job application failure scenario (multiple interview rounds ending in rejection in the final round) prior to the start of Experiment 2 to induce mild frustration. They then sequentially experienced 16 stimuli: four single-modal audio files in the no-vibration condition, delivered via an external speaker with no haptic feedback; 12 multimodal stimuli in the vibration condition, combining audio with different haptic path patterns (as detailed in Section 2.3). To ensure consistency and reduce cognitive load, the spatial haptic parameter was limited to the small-area condition, and the external speaker was calibrated for audio quality. Each stimulus was separated by a 25-second interval, allowing participants sufficient time to complete their ratings and return to baseline states. Participants rated each stimulus using a 9-point Likert scale [12] in three dimensions:

- Trust: Did the system align with your emotions and make you feel trust?
- Support Effectiveness: Did the system encourage and support you?
- Empowerment: Did the system make you feel more capable of facing challenges?

These dimensions were chosen to comprehensively evaluate cognitive aspects of the impact of the system, focusing on its ability to foster connection, provide meaningful support, and enhance users' self-efficacy [4]. After the experiment, participants participated in a semi-structured interview to share their impressions of the stimuli and provided suggestions for improvement.

4 RESULTS

4.1 Multimodal Haptic and Music Stimuli: Effects on Emotional Experience

A three-factor repeated measures ANOVA was conducted to examine the effects of music type, haptic vibration path and area on pleasure and arousal ratings, with Greenhouse-Geisser correction applied when necessary.

4.1.1 Evaluation of the Arousal Dimension. The main effect of music type was significant ($F(3, 33) = 4.69, p = 0.007$). LSD post-hoc tests indicated that HAHV music elicited significantly higher arousal ratings than other types, while HALV, LALV, and LAHV showed no significant differences. As shown in Fig.2a, this pattern is consistent across all path types, with HAHV exhibiting the highest mean ratings, whereas the other three music types display similar distributions. The main effect of path type was also significant ($F(2, 22) = 4.48, p = 0.023$). As illustrated in Fig.2a, arousal ratings

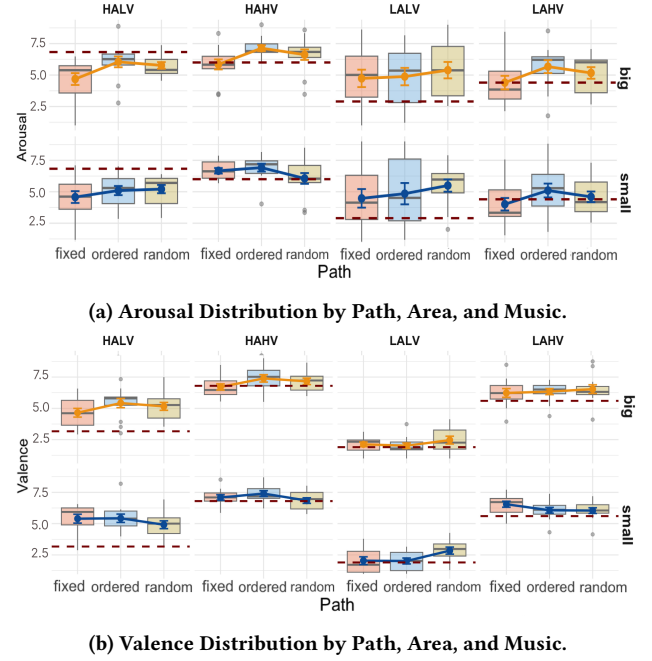


Figure 2: The Influence of Music, Path, and Area on Emotional Dimensions. In subfigure (a), the red dashed line represents the baseline arousal of each music type; in subfigure (b), the red dashed line represents the baseline valence of each music type.

in the “fixed” path type were generally lower, with its mean values consistently below those of the “ordered” path and “random” path. No significant difference was observed between the latter two. Participants described the “fixed” path as “focusing and calming”, suggesting that its rhythmic stability may reduce arousal. As shown in Fig.2a, apart from the LALV music type, the arousal levels for the “ordered” path were generally slightly higher than for the “random” path. Recording the interview, regular haptic patterns were perceived as “like flowing water, which can better enhance the atmosphere”. In contrast, the unpredictability of the “random” path type is often labeled “distracting”. However, under the LALV music condition, participants reported that “the added vibration made me feel somewhat excited and happy.”

4.1.2 Evaluation of the Valence Dimension. The main effect of music type on valence was significant ($F(3, 33) = 128.11, p < 0.001$). LSD post-hoc tests revealed significant differences between all music types, with high-valence music receiving significantly higher ratings than low-valence types. This trend can also be visually distinguished through the differences in mean values among the music groups in Fig.2b. As noted by most participants during the interviews, the perceived pleasantness of the multimodal stimuli was primarily driven by the type of music. The main effect of haptic path types was not significant. However, the interaction effect between haptic path types and music types was significant ($p = 0.041$). Figure 2b shows that the “random” path reduced negative

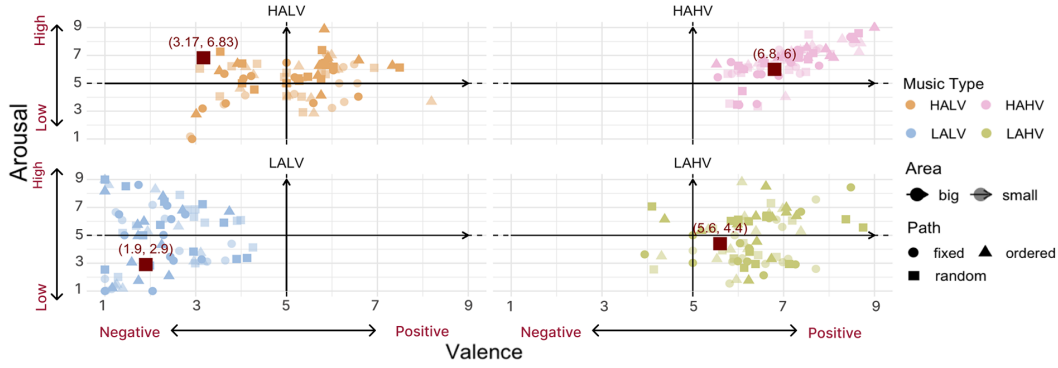


Figure 3: The distribution of all data on the arousal-valence two-dimensional coordinate system.

emotions under LALV music, while the “ordered” path enhanced positive emotions with high-arousal music.

4.1.3 Analysis Based on Displacement. To assess the impact of multimodal stimuli on the original emotional expression of music, we analyzed Euclidean distances and distribution trends in the arousal-valence coordinate plane. This quantified the magnitude and direction of emotional changes from integrating haptic parameters with music, highlighting the interaction between haptic and musical emotions and revealing the effects of multimodal stimuli on emotional expression. Figure 3 illustrates the distribution of all 288 data points in the Arousal-Valence quadrant. Combined with the Euclidean distance visualization in Fig.4, the results reveal differences in Arousal and Valence shifts across multimodal stimuli for different music types.

Low-Arousal Music. By analyzing the lower two axes of Fig.3, data points predominantly shift upward and to the right, with approximately 50% crossing quadrants. This suggests that haptic feedback may enhance emotional arousal. P5 noted, “*The calm music, combined with vibration, made me feel empowered.*”

Low-Valence Music. Figure 4 shows that shift distances of low-valence music were generally larger, particularly under fixed and random conditions. Further analysis of the upper-left coordinate system in Fig.3 reveals that HALV music predominantly shifted to the right, with about half of the data points crossing quadrants. This suggests that low-valence music may evoke distinct emotional shift patterns. Some participants noted that “*vibration seems to guide emotions toward a more positive direction,*” while others mentioned that “*negative music combined with vibration makes emotional expression more complex.*”

High-Valence Music. Figure 4 shows that low-valence music had smaller shift distances, and HAHV music exhibited the smallest shifts across all paths. The upper-right coordinate system in Fig.3 illustrates that under the HAHV condition, most data points clustered near the initial coordinates. Some participants commented, “*When vibration is added to joyful music, it seems to enhance the existing emotion rather than introduce new changes.*” As shown in Fig.4, the Euclidean distance for HAHV music remained minimal regardless of haptic area types or path types.

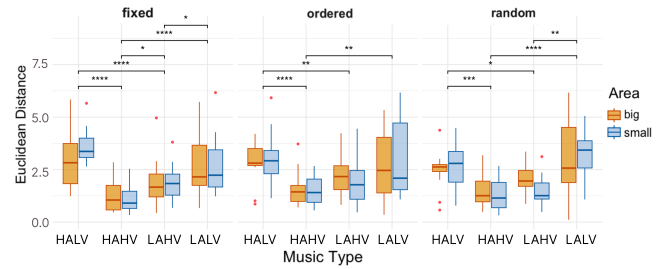


Figure 4: Comparison of emotion offset distance by spatial haptic parameters. (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$).

4.2 Effects of Single-Voice and Multimodal Stimuli in Encouragement Scenarios

The three-factor repeated measures ANOVA results revealed a significant main effect of path pattern on ratings of empowerment ($F(3, 48) = 14.26, p < 0.001, \eta^2 = 0.249$), support ($F(3, 48) = 15.20, p < 0.001, \eta^2 = 0.258$), and trust ($F(3, 48) = 12.15, p < 0.001, \eta^2 = 0.223$). In contrast, language content, voice gender, and their interactions showed no significant effects. The visualization of these results is presented in Fig.5. Tukey HSD post hoc results and the rating trends in Fig.5 indicate that significant differences primarily arise from the no-vibration condition compared to vibration conditions, and from the fixed path condition compared to the ordered and random path conditions.

Comparison Examined Conditions with and without Vibration: As shown in Fig.5, the null (no-vibration) condition consistently received the lowest ratings across all three dimensions. Correlation analysis ($r > 0.8, p < 0.01$) further demonstrated consistency in the cognitive support provided by path patterns across dimensions. The feedback from the interviews highlighted the realism introduced by vibration. For example, P5 noted, “*It felt like Hugvie was speaking, recreating the vibrations of a conversation.*” P7 commented, “*Without vibration, there was a sense of distance, as if the topic wasn’t relevant to me.*” P11 added, “*Being encouraged by the system made me feel warm and even a bit charmed.*”

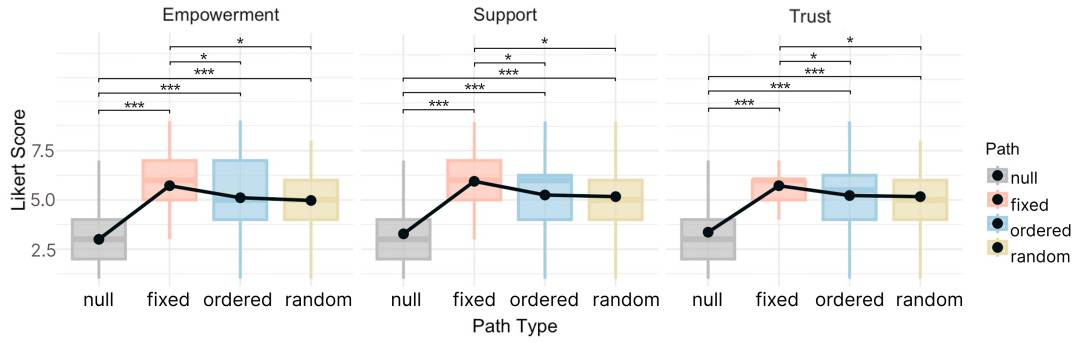


Figure 5: Rating differences across three dimensions under haptic vibration conditions. (* $p < 0.05$, *** $p < 0.001$).

Haptic Path Comparison: “Fixed” path performed significantly better than other paths, with ordered path slightly outperforming random path, as shown in Fig.5, where the “fixed” path consistently received the highest mean ratings across all dimensions. Although Tukey HSD tests found no significant difference between “ordered” and “random” paths ($p > 0.05$), Fig.5 shows a slight upward trend for “ordered path, suggesting a marginal preference. Participants supported this in interviews: P9 shared, “Fixed vibrations feel calming, like a heartbeat,” and P10 added, “Fixed or directional vibrations transmit a sense of energy.”

5 DISCUSSION AND FUTURE WORKS

5.1 Findings

The study provides preliminary insights into the potential role of multimodal stimuli in empathy expression. First, Music-driven multimodal stimuli had stronger effects on valence than arousal, likely due to increased arousal from haptic vibrations, reducing differentiation between music types. Second, Ordered paths, providing stable haptic stimulation, generally elicited higher arousal than fixed positions, since the dynamic sensation of position changes in ordered paths. Additionally, haptic patterns that are both vibratory and stable are more favored in cognitive support scenarios. Participants reported that the multimodal vibrations were subjectively perceived as resembling “vocal cord vibrations,” evoking a sense of hugging a real person. This perception of natural rhythm may enhance users’ trust in the system. Future research could further explore the relationship between vibration rhythms and trust. While random paths elicited positive experiences under certain conditions (e.g., P10 noted, “Vibrations during negative music made me feel excited and happy”), half of the participants reported increased cognitive load. Lastly, larger vibration areas, though not significant in ratings, were noted in interviews as effective for creating atmosphere.

5.2 Limitations and Future Work

For future work, we aim to include participants from diverse backgrounds to improve generalizability. Auditory stimuli should feature more varied music and linguistic content tailored to cognitive support. Haptic design should balance predictability and variation to explore the role of dynamic paths in empathy and examine how vibration rhythms influence trust. Expanding vibration area options can clarify their impact on emotions.

Moreover, while this study primarily focuses on vibration as the haptic stimuli due to its natural integration with auditory stimuli, future research will explore the role of other haptic stimuli, such as pressure and temperature, in empathy expression by varying spatial parameters. In particular, we aim to investigate how these haptic modalities influence different emotional states and their complementarity with vibration. Furthermore, future studies will systematically evaluate the impact of different haptic modalities on users’ emotional perception and empathy experience to determine their optimal applications in various interaction contexts.

Also, we need evaluate the effects of stimuli when users are in different emotional states. To enhance automation and natural interaction, real-time emotion recognition using physiological signals will be introduced, enabling dynamic adjustments to haptic parameters and audio selection for personalized feedback.

6 CONCLUSION

In this study, we explored the impact of spatial haptic parameters and auditory cues on empathy expression in HRI systems. To achieve this, we developed a huggable robot system capable of delivering multimodal stimuli through vibration speakers. Two experiments evaluated the effects of haptic-music and haptic-speech combinations. Results revealed that haptic path parameters influenced arousal, while music significantly affected valence, and synergistic effects emerged under specific conditions. In cognitive support, vibration combined with speech outperformed voice-only stimuli, while random haptic paths performed worse. We believe these findings provide valuable insights for optimizing multimodal design to enhance empathy expression in HRI systems.

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